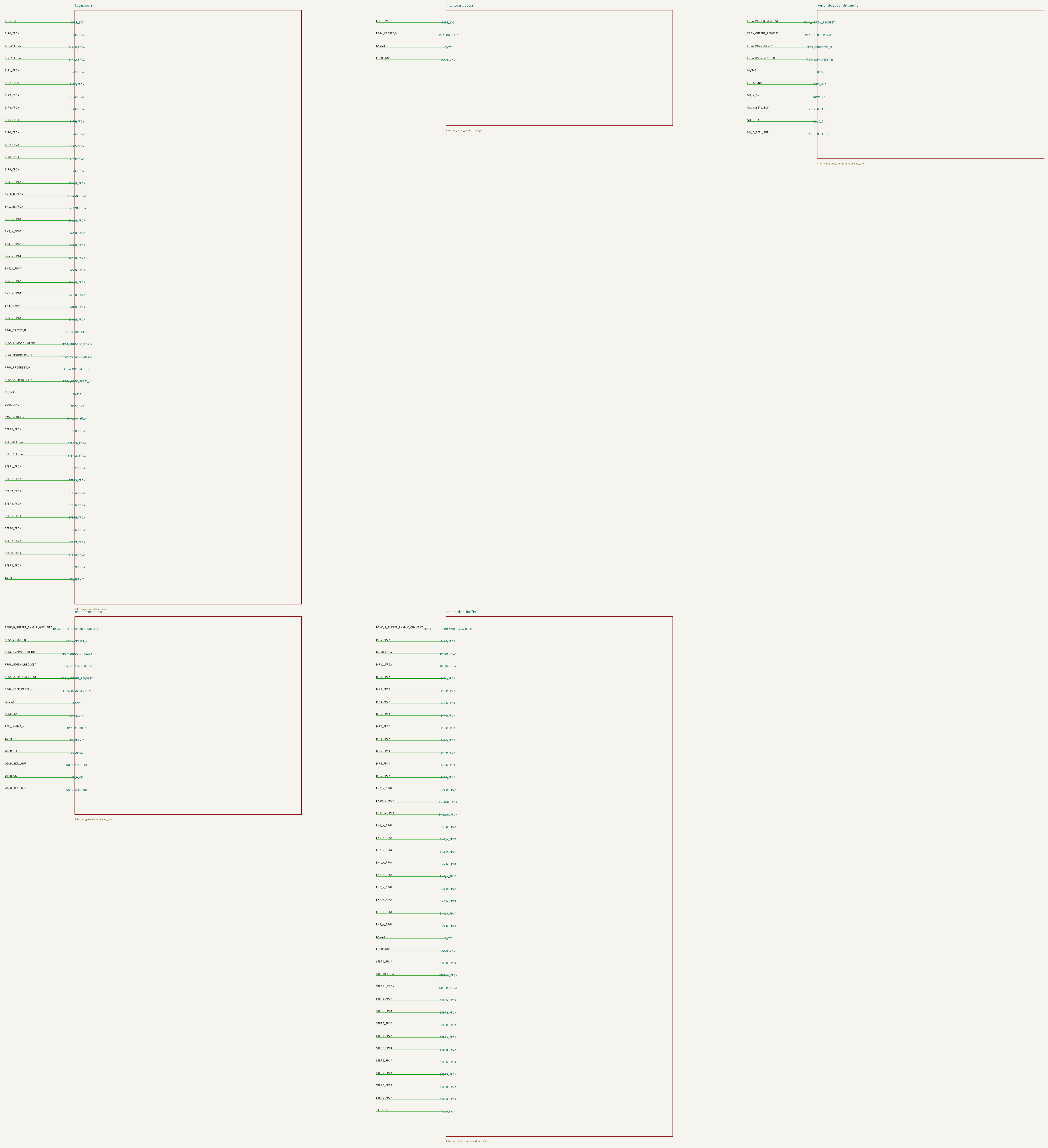
PROPOSED IS-2 MC CIRCUIT CANDIDATE: FPGA, autonomous local rails, conditioned dual watchdogs, permission/OPI/trip latches and 12-motor Ioff buffers.
38 motor/TX paths use actual Ioff buffers; other Ioff paths, presence filters, identity, slot-status register, flash restore and connectors remain missing.
MC_RAILS_OK uses qualified destination-rail FPGA_CRESET_N, not MC_RAILS_OK_5V. CDONE qualification uses delayed FPGA_USER_RESET_N; timing is unqualified.
Legacy progress ball is proposed as FPGA_PROGRESS_M; A2 is the independent RAW_INHIBIT_N input. Reserved enable balls now connect to EN_N buffers; vendor fit, CNC1 pulse generation and remaining GPIO allocation are open.
Bank-B qualified OE is the same physical net as OPI_BOE clear. Receiving cards must pull EN_N high from their own powered rail; those receivers remain separate unfinished circuits.
Native schematic ERC is distinct from PCB DRC, vendor FPGA fit and measurements. NOT FOR MANUFACTURE.


fpga_core
CORE_1V2
DIR0_FPGA
DIR10_FPGA
DIR11_FPGA
DIR1_FPGA
DIR2_FPGA
DIR3_FPGA
DIR4_FPGA
DIR5_FPGA
DIR6_FPGA
DIR7_FPGA
DIR8_FPGA
DIR9_FPGA
EN0_N_FPGA
EN10_N_FPGA
EN11_N_FPGA
EN1_N_FPGA
EN2_N_FPGA
EN3_N_FPGA
EN4_N_FPGA
EN5_N_FPGA
EN6_N_FPGA
EN7_N_FPGA
EN8_N_FPGA
EN9_N_FPGA
FPGA_CRESET_N
FPGA_ENDPOINT_READY
FPGA_MOTION_REQUEST
FPGA_PROGRESS_M
FPGA_USER_RESET_N
IO_3V3
LOGIC_GND
RAW_INHIBIT_N
STEP0_FPGA
STEP10_FPGA
STEP11_FPGA
STEP1_FPGA
STEP2_FPGA
STEP3_FPGA
STEP4_FPGA
STEP5_FPGA
STEP6_FPGA
STEP7_FPGA
STEP8_FPGA
STEP9_FPGA
TX_PERMIT
File: fpga_core.kicad_sch
mc_local_power
CORE_1V2
FPGA_CRESET_N
IO_3V3
LOGIC_GND
File: mc_local_power.kicad_sch
watchdog_conditioning
FPGA_MOTION_REQUEST
FPGA_OUTPUT_REQUEST
FPGA_PROGRESS_M
FPGA_USER_RESET_N
IO_3V3
LOGIC_GND
WD_M_OK
WD_M_SET1_BUF
WD_O_OK
WD_O_SET1_BUF
File: watchdog_conditioning.kicad_sch
mc_permission
BANK_B_BUFFER_ENABLE_QUALIFIED
FPGA_CRESET_N
FPGA_ENDPOINT_READY
FPGA_MOTION_REQUEST
FPGA_OUTPUT_REQUEST
FPGA_USER_RESET_N
IO_3V3
LOGIC_GND
RAW_INHIBIT_N
TX_PERMIT
WD_M_OK
WD_M_SET1_BUF
WD_O_OK
WD_O_SET1_BUF
File: mc_permission.kicad_sch
mc_motor_buffers
BANK_B_BUFFER_ENABLE_QUALIFIED
DIR0_FPGA
DIR10_FPGA
DIR11_FPGA
DIR1_FPGA
DIR2_FPGA
DIR3_FPGA
DIR4_FPGA
DIR5_FPGA
DIR6_FPGA
DIR7_FPGA
DIR8_FPGA
DIR9_FPGA
EN0_N_FPGA
EN10_N_FPGA
EN11_N_FPGA
EN1_N_FPGA
EN2_N_FPGA
EN3_N_FPGA
EN4_N_FPGA
EN5_N_FPGA
EN6_N_FPGA
EN7_N_FPGA
EN8_N_FPGA
EN9_N_FPGA
IO_3V3
LOGIC_GND
STEP0_FPGA
STEP10_FPGA
STEP11_FPGA
STEP1_FPGA
STEP2_FPGA
STEP3_FPGA
STEP4_FPGA
STEP5_FPGA
STEP6_FPGA
STEP7_FPGA
STEP8_FPGA
STEP9_FPGA
TX_PERMIT
File: mc_motor_buffers.kicad_sch

Sheet: /
File: motion_card.kicad_sch

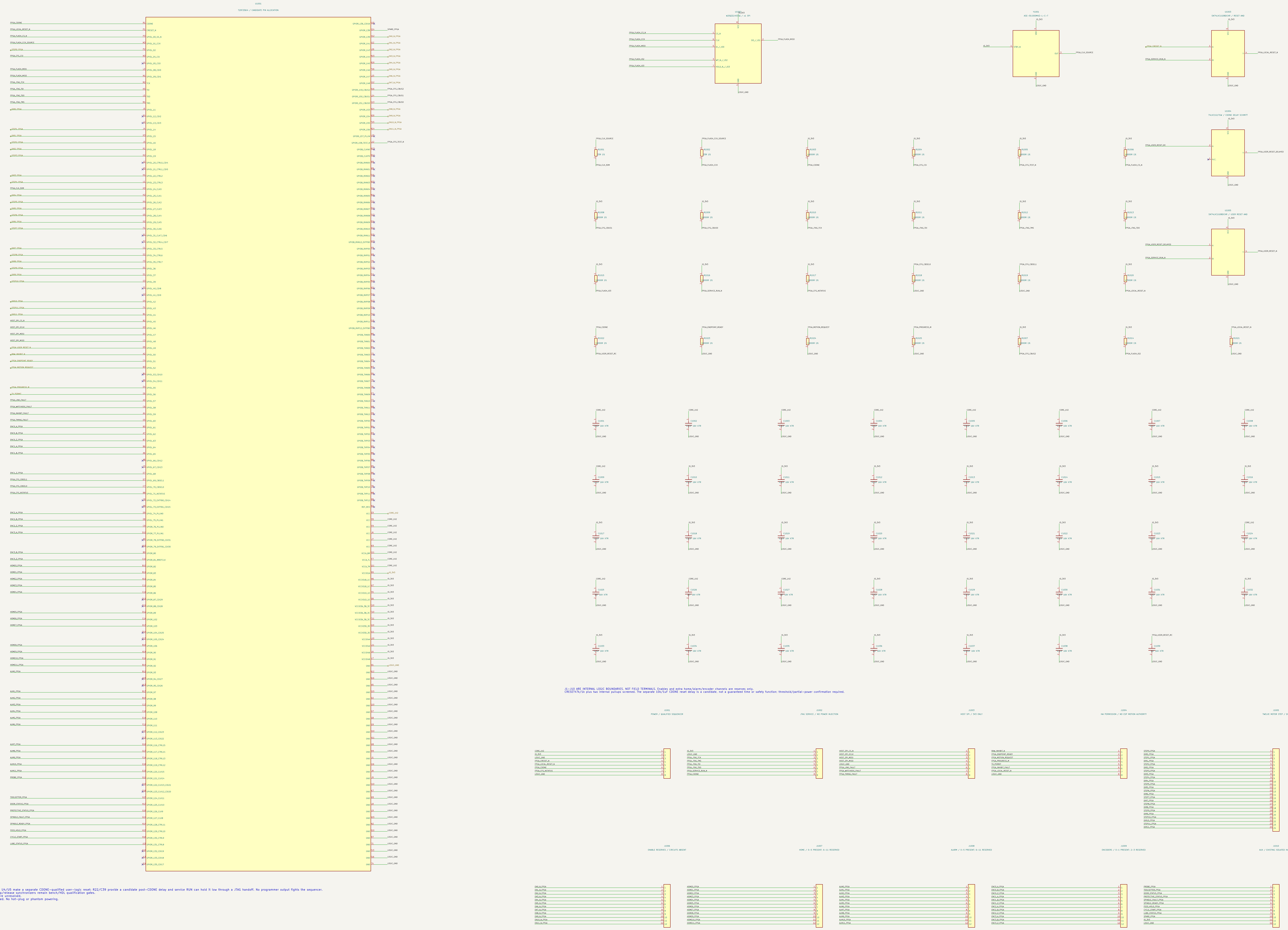
Title: Motion card circuit candidate

Size: A0
Date: 2026-09-30
Rev: r0.4-proposed-IS2
KiCad E.D.A. 10.0.4
Id: 1/6

T20F256 FPGA CORE / MINIMUM 8 MOTOR CHANNELS, 12–CHANNEL PIN CAPACITY CANDIDATE

Exact 256 balls: Efinix T20–PINOUT worksheet T20F256. Pin allocation is a board candidate, NOT vendor fit or routed timing.
All VCCIO=3.3V. All VCC/VCCA=1.2V. No 5V–tolerant FPGA connection. No native LVDS: REF_RES intentionally NC.

One 100nF per supply ball; bulk per physical supply group. Decoupling is a layout–study allowance, not a PDN proof.
Active SPI x1 straps111; CSI/TEST high. Flash bus reserved from user RTL. CCK damping33R is provisional.
CDONE is open–drain with10k pullup; status does not grant motion permission. Configuration clock is internal, not Y1.


U1001
T20F256I4 / CANDIDATE PIN ALLOCATION
U1002
W25Q32JVSSIQ / x1 SPI
Y1001
ASE–50.000MHZ–L–C–T
U1003
SN74LVC1G08DCKR / RESET AND
U1004
74LVC1G17GW / CDONE DELAY SCHMITT
U1005
SN74LVC1G08DCKR / USER RESET AND
J1001
POWER / QUALIFIED SEQUENCER
J1002
JTAG SERVICE / NO POWER INJECTION
J1003
HOST SPI / 3V3 ONLY
J1004
HW PERMISSION / NO ESP MOTION AUTHORITY
J1005
TWELVE MOTOR STEP / DIR
J1006
ENABLE RESERVES / CIRCUITS ABSENT
J1007
HOME / 0–5 PRESENT; 6–11 RESERVED
J1008
ALARM / 0–5 PRESENT; 6–11 RESERVED
J1009
ENCODERS / 0–1 PRESENT; 2–3 RESERVED
J1010
AUX / EXISTING ISOLATED INPUTS

J1–J10 ARE INTERNAL LOGIC BOUNDARIES, NOT FIELD TERMINALS. Enables and extra home/alarm/encoder channels are reserves only.
CRESET47k/1k plus two internal pullups screened. The separate 10k/1uF CDONE reset delay is a candidate, not a guaranteed time or safety function; threshold/partial–power confirmation required.

RESET: U3 combines qualified upstream reset with service RUN for configuration reset only. U4/U5 make a separate CDONE–qualified user–logic reset; R22/C39 provide a candidate post–CDONE delay and service RUN can hold it low through a JTAG handoff. No programmer output fights the sequencer.
User–logic reset release threshold, RC tolerance, discharge time, tUSER handoff and startup/release synchronizers remain bench/HDL qualification gates.
J2 is a logical service boundary: actual keyed connector, adapter, cable loading and ESD are unresolved.
JTAG programming requires a deliberate reset cycle, valid rails and machine outputs inhibited. No hot–plug or phantom powering.

NC crosses are genuine unused GPIO/config outputs and REF_RES only; every supply/GND ball is connected.
Boundary symbols are not released connectors. No footprint approval, PCB, DRC, bitstream, qualification or fabrication files.

Sheet: /fpga_core/
File: fpga_core.kicad_sch
Title:
Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 2/6

MC LOCAL POWER – IS–2 HARDWARE PROPOSAL / NOT MANUFACTURING READY

CORE_1V2: 1.2 V / 0.8 A reservation; 1 MHz forced PWM

IO_3V3: 3.3 V / 0.15 A reservation; 2.5 MHz forced PWM

U2001
TPS62901RPJR
VIN
EN
MODE/S–CONF
SS/TR
SW
PG
VOS
FB/VSET
GND
L2001
2.2uH
CORE_5V
CORE_1V2_SW
CORE_1V2_PG_OD
CORE_1V2
LOGIC_GND
U2002
TPS62901RPJR
L2002
1uH
IO_3V3_SW
IO_3V3_PG_OD
IO_3V3
U2003
LTC2962HUD#TRPBF
VCC
V1
V2
V3
V4
MR
RDIS
DVCC
RST
REF
PG1
PG2
PG3
PG4
RT
EP
CORE_5V_UV_SENSE
CORE_5V_OV_SENSE
MC_SEQ_5V_OK
MC_SEQ_REF1
U2004
LTC2962HUD#TRPBF
U2005
LTC2962HUD#TRPBF
MC_RAILS_OK_5V
U2006
SN74LVC1G17DBVR
NC
FPGA_CRESET_N
B2001
MC POWER REVIEW BOUNDARY
CORE_5V
LOGIC_GND
CORE_1V2
IO_3V3
FPGA_CRESET_N
MC_RAILS_OK_5V
CORE_1V2_PG_OD
IO_3V3_PG_OD
LOGIC_GND
pwr

5 V window releases core; core window releases IO; IO window releases FPGA reset.
Each supervisor release: 160–240 ms. Raw MC_RAILS_OK_5V is not a 3.3 V signal.
FPGA_CRESET_N is the buffered rails–good term; combine other reset requests upstream.
Regulator PG pins are diagnostic only; no PG–to–EN cascade. Physical connector is external.

Actual rail limits retained: 5 V 4.85–5.15; T20 core 1.15–1.25; IO 3.20–3.40 V.
Core thresholds include initial/temp plus 400ppm drift; historical tight ledger is distinct.
IO uses 2.5 MHz because existing FPGA decoupling exceeds the 1 MHz capacitance limit.
Proposed MC slot: 0.50 A steady / 0.80 A downstream startup; input charging separate.
Arbitrary partial power, assembled stress, brownout slew, shutdown and timing unproved.

Sheet: /mc_local_power/
File: mc_local_power.kicad_sch
Title: MC local regulators and independent window sequencing
Size: A1 Date: 2026–09–30 **Rev: r0.1 / IS–2 proposal**
KiCad E.D.A. 10.0.4 Id: 3/6

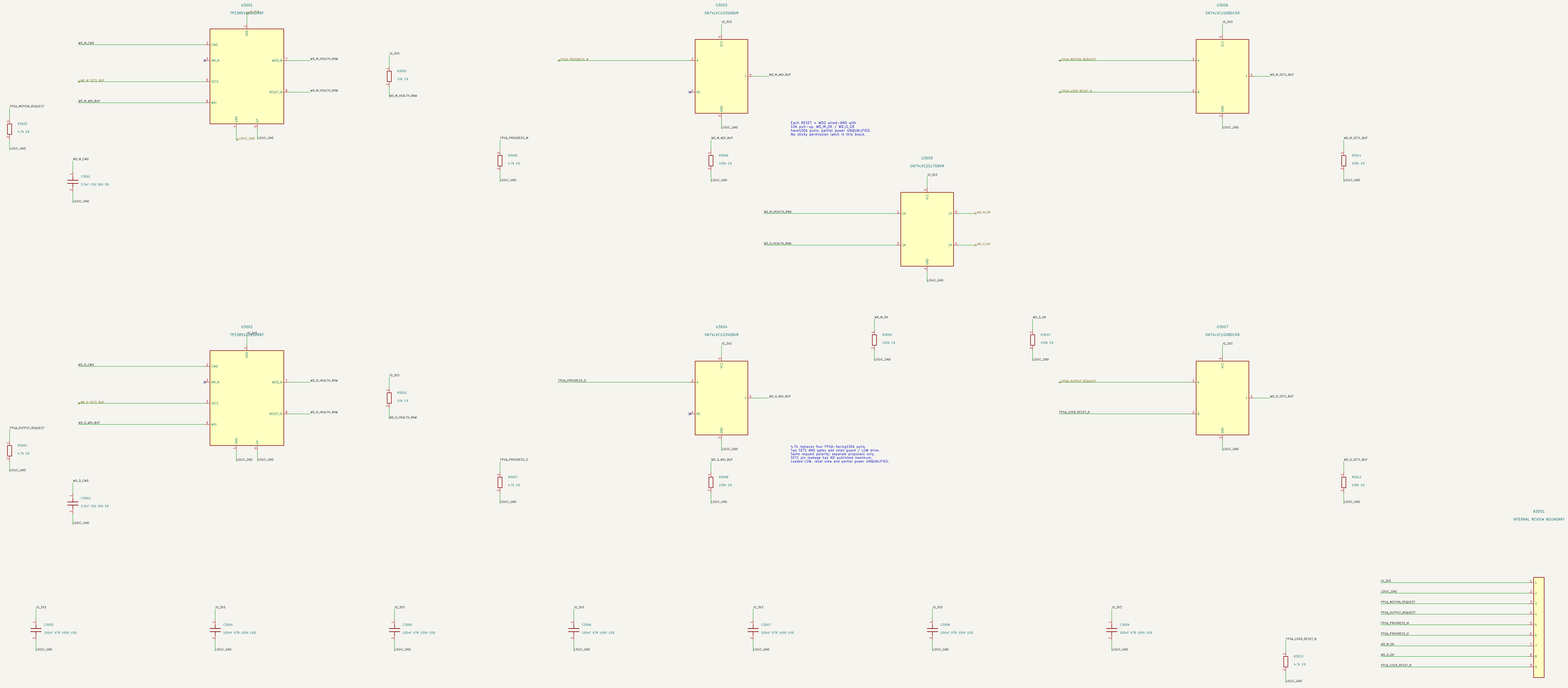
IS–2 PROPOSAL: DUAL WATCHDOG STARTUP CONDITIONING – SEPARATE REVIEW CANDIDATE

NOT CONNECTED TO THE ASSEMBLY. No complete permission chain, latches, field outputs, PCB or physical qualification.

Separate timers / shared IO_3V3 and LOGIC_GND / G18 threshold is not a 3.3V power–good qualification / all inputs INTERNAL logic.

4.7k FPGA–facing pulls; SET1 = request AND FPGA_USER_RESET_N. Active–HIGH polarity / 1us pulse contract unchanged. MR open.

WDI: one 1us high pulse, idle LOW. FALLING edges feed timer.
Old toggle–per–UPDATE gateware is INCOMPATIBLE.
3.3nF C0G: calculated timing, never a measured20ms proof.


U3001
TPS3851G18SDRBT
IO_3V3
VDD
CWD
MR_N
SET1
WDI
WDO_N
RESET_N
GND
EP
WD_M_CWD
WD_M_SET1_BUF
WD_M_WDI_BUF
WD_M_HEALTH_RAW
LOGIC_GND
R3001
10k 1%
R3003
4.7k 1%
FPGA_MOTION_REQUEST
C3001
3.3nF C0G 50V 5%
U3002
TPS3851G18SDRBT
WD_O_CWD
WD_O_SET1_BUF
WD_O_WDI_BUF
WD_O_HEALTH_RAW
R3002
10k 1%
R3004
4.7k 1%
FPGA_OUTPUT_REQUEST
C3002
3.3nF C0G 50V 5%
U3003
SN74LVC1G34DBVR
FPGA_PROGRESS_M
A
NC
Y
VCC
R3005
4.7k 1%
R3006
100k 1%
U3004
SN74LVC1G34DBVR
FPGA_PROGRESS_O
WD_O_WDI_BUF
R3007
4.7k 1%
R3008
100k 1%
U3005
SN74LVC2G17DBVR
1A
2A
1Y
2Y
WD_M_OK
WD_O_OK
R3009
100k 1%
R3010
100k 1%
U3006
SN74LVC1G08DCKR
FPGA_USER_RESET_N
B
R3011
100k 1%
U3007
SN74LVC1G08DCKR
WD_O_SET1_BUF
R3012
100k 1%
B3001
INTERNAL REVIEW BOUNDARY
IO_3V3
LOGIC_GND
FPGA_MOTION_REQUEST
FPGA_OUTPUT_REQUEST
FPGA_PROGRESS_M
FPGA_PROGRESS_O
WD_M_OK
WD_O_OK
FPGA_USER_RESET_N
R3013
4.7k 1%
C3003
C3004
C3005
C3006
C3007
C3008
C3009
100nF X7R 100V 10%

Each RESET + WDO wired–AND with
10k pull–up. WD_M_OK / WD_O_OK
have100k pulls; partial power UNQUALIFIED.
No sticky permission latch in this block.

4.7k replaces four FPGA–facing100k pulls.
Two SET1 AND gates add reset guard / LOW drive.
Same request polarity; separate proposals only.
SET1 pin leakage has NO published maximum.
Loaded LOW, reset slew and partial power UNQUALIFIED.

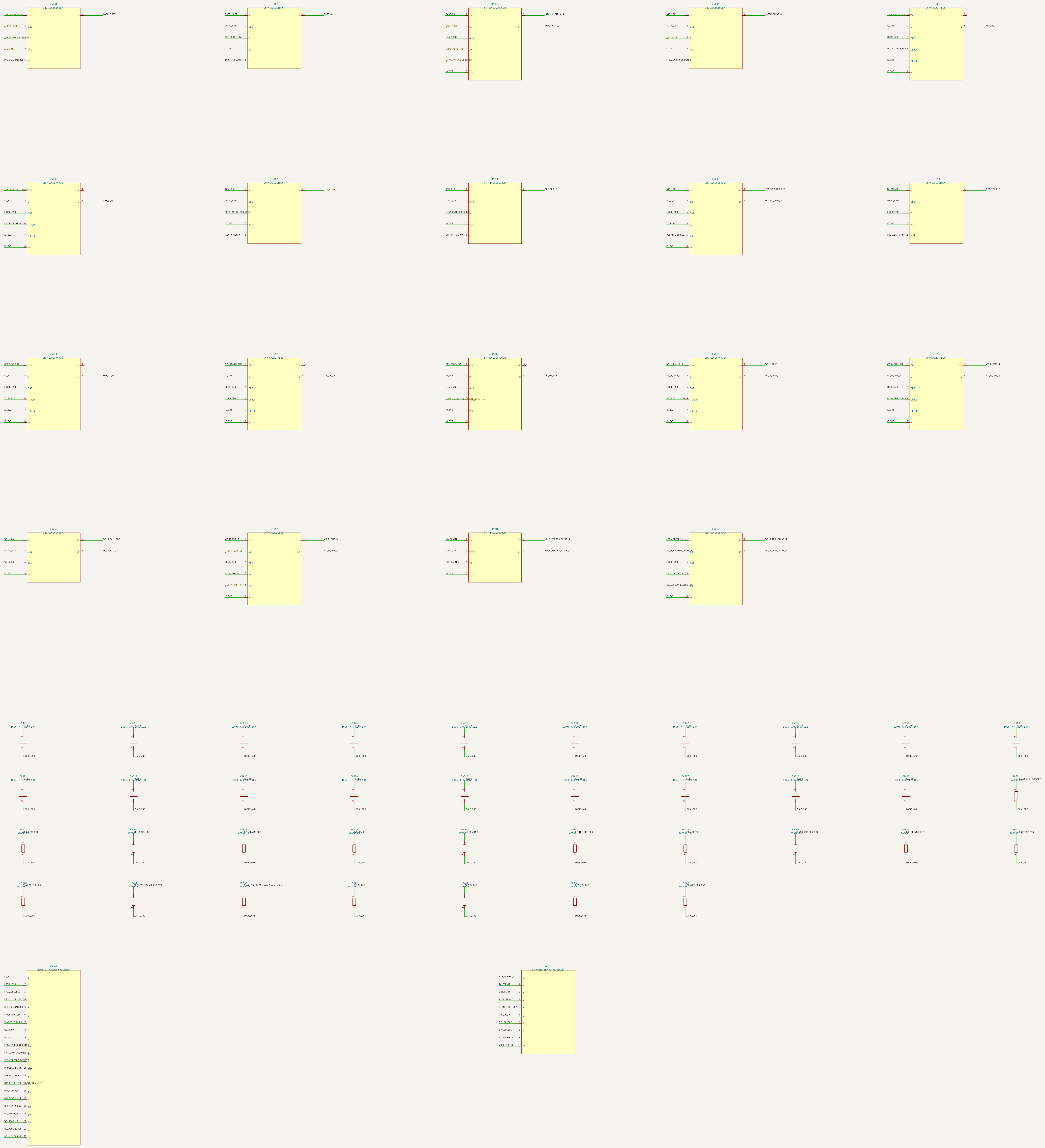
DRB copper–pad candidate only: paste/stencil, thermal vias, CWD guarding/cleanliness/leakage, routing, partial power and brownout remain unqualified.

Sheet: /watchdog_conditioning/
File: watchdog_conditioning.kicad_sch

Title: IS–2 proposed watchdog conditioning – review only

Size: A1	Date: 2026–09–30	**Rev: r0.2–proposal**
KiCad E.D.A. 10.0.4		Id: 4/6

MC PERMISSION / STICKY INTERRUPTION CORE — UPSTREAM QUALIFICATION AND FIELD DRIVERS REMAIN SEPARATE



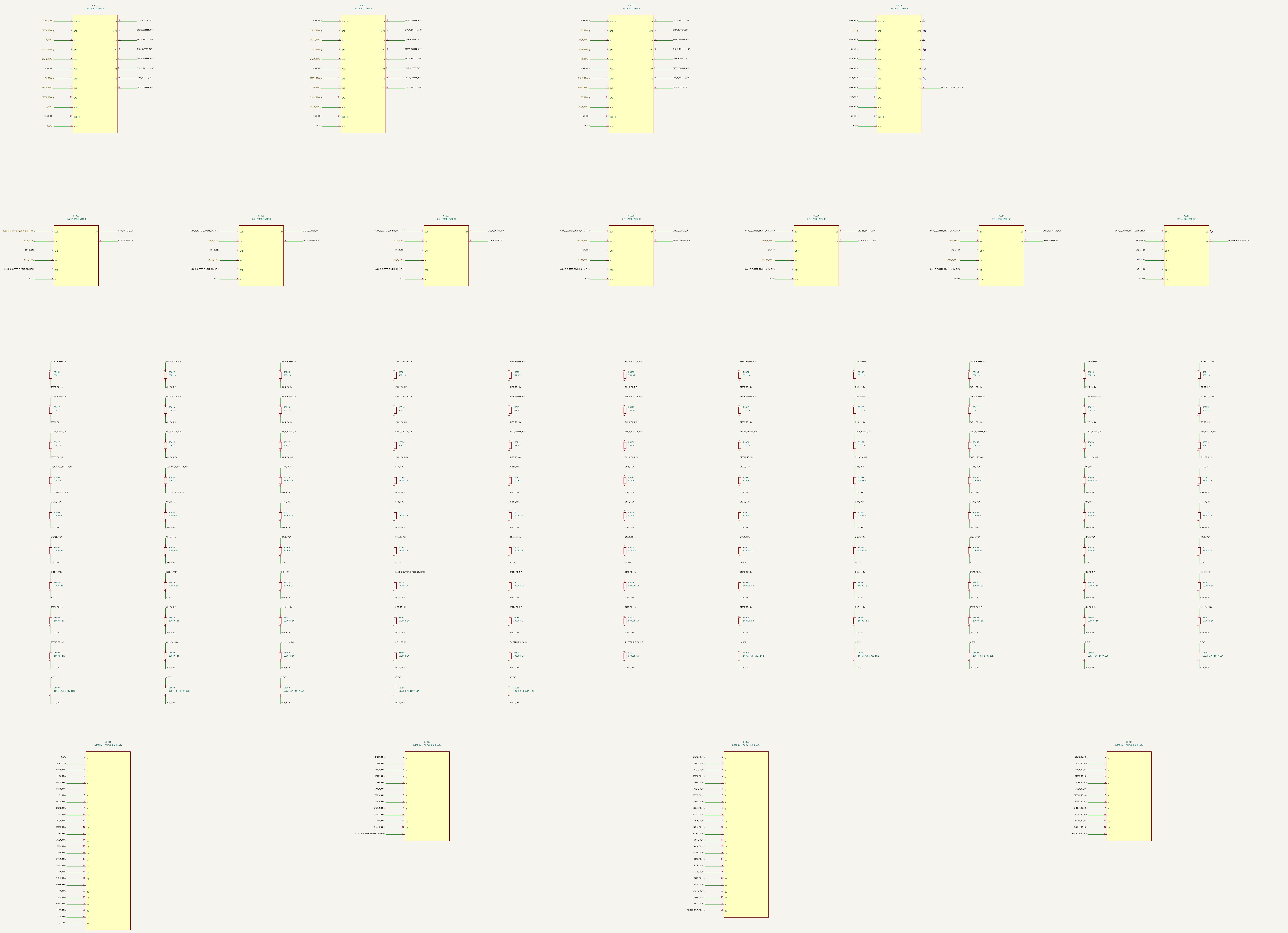
No manufacture/physical timing claim. IS−1 arm/OPI equations; 4.7k GPIO startup pulls need proposed IS−2 reconciliation.
WD_TRIP_N: a WD_OK FALL captures actual timer SET1; sticky Q feedback, power reset or explicit WD_REARM pulse clears the record.
B1/B2 are logical boundaries, not connectors. Trip clear protocol, loaded edge timing and upstream qualified terms remain open.

Sheet: /mc_permission/
File: mc_permission.kicad_sch

Title: Motion−card permission core

Size: A0	Date: 2026−09−30	**Rev: r0.2**
KiCad E.D.A. 10.0.4		Id: 5/6

MC MOTOR / TX Ioff: 8 STANDARD +4 EXPANSION; B1–B4 ARE LOGICAL BOUNDARIES, NOT CONNECTORS


U5001
SN74LVC244APWR
U5002
SN74LVC244APWR
U5003
SN74LVC244APWR
U5004
SN74LVC244APWR
U5005
SN74LVC2G126DCUR
U5006
SN74LVC2G126DCUR
U5007
SN74LVC2G126DCUR
U5008
SN74LVC2G126DCUR
U5009
SN74LVC2G126DCUR
U5010
SN74LVC2G126DCUR
U5011
SN74LVC2G126DCUR
B5001
INTERNAL LOGICAL BOUNDARY
B5002
INTERNAL LOGICAL BOUNDARY
B5003
INTERNAL LOGICAL BOUNDARY
B5004
INTERNAL LOGICAL BOUNDARY

BankA244A OEs tiedLOW(active). BankB2G126 used OEs share BANK_B_BUFFER_ENABLE_QUALIFIED with parentOPI_BOE clear.
EN_N outgoing pull–ups belong on poweredMIA/MX4 receiver rails; NEVER on deadMC rail. 33R series are unqualified SI candidates.
Not completeMC; no routing, targetfit, loadedtiming or manufacture claim.

Sheet: /mc_motor_buffers/
File: mc_motor_buffers.kicad_sch
Title: Motion–card motor Ioff boundaries
Size: A0 Date: 2026–09–30 **Rev: r0.1**
KiCad E.D.A. 10.0.4 Id: 6/6